2nd Midterm · Unit 6 · Heat and wave equations by separation · 5 points · 45 min · Standard exam level

P14 — Vibrating string with fixed ends started from the rest position with a parabolic initial velocity: modes, coefficients from the velocity, d'Alembert check at the midpoint, energy of each mode and periodicity

A string of length L, linear density ρ and tension T_0 is fixed at $x = 0$ and $x = L$. Its small transverse displacement $u(x,t)$ satisfies

$$u_{tt} = c^2 u_{xx}, \qquad 0 < x < L,\ t > 0, \qquad c^2 = \frac{T_0}{\rho}.$$

At $t = 0$ the string lies along its rest position and is struck by a wide hammer that gives it the velocity profile

$$u(x,0) = 0, \qquad u_t(x,0) = g(x) = \frac{4V}{L^2}\, x(L - x),$$

where $V > 0$ is the speed given to the midpoint.

(a) Separate variables, discuss the sign of the separation constant and find the normal modes compatible with the fixed ends and with $u(x,0) = 0$.

(b) Obtain $u(x,t)$ as a series, with the coefficients in closed form.

(c) For $0 \le t \le L/(2c)$ use d'Alembert's formula to obtain the displacement of the midpoint, $u(L/2,t)$, in closed form. Check it against the series at $t = L/(2c)$ and deduce $\sum_{n\ \mathrm{odd}} n^{-4}$.

(d) The energy of the string is $E(t) = \displaystyle\int_0^L \Big(\frac{\rho}{2} u_t^2 + \frac{T_0}{2} u_x^2\Big)\,dx$. Compute the energy carried by each mode, check that their sum is the energy given by the hammer, and give the fraction carried by the fundamental. Show that the motion is periodic in time and describe the string at $t = L/(2c)$ and $t = L/c$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(x,t)$ on $0 \le x \le L, t \ge 0$; small transverse vibrations, uniform string, no damping, no gravity.
- Operator: linear, second order, hyperbolic, homogeneous, with homogeneous Dirichlet conditions $u(0,t) = u(L,t) = 0$.
- Data: L [m], c [m/s], V [m/s], ρ [kg/m], $T_0 = \rho c^2$ [N]. Two initial conditions are needed (second order in t): zero displacement, non-zero velocity.

Phase 2. Canonical formulation

$$u_{tt} = c^2 u_{xx}, \quad u(0,t) = u(L,t) = 0, \quad u(x,0) = 0, \quad u_t(x,0) = g(x)$$

Eigenvalue problem in x: $X'' + \lambda X = 0$, $X(0) = X(L) = 0$. In t: $T'' + \lambda c^2 T = 0$, with $T(0) = 0$ from the zero initial displacement.

Part (a) · 1.25 points

Separation and normal modes

Step 1. Separation

$XT'' = c^2 X'' T$, so $\dfrac{T''}{c^2 T} = \dfrac{X''}{X} = -\lambda$ (each side depends on one variable only). Fixed ends: $X(0)T(t) = 0$ for all t, so $X(0) = 0$, and $X(L) = 0$.

Step 2. Case lambda < 0

$\lambda = -\mu^2$: $X = a\cosh \mu x + b \sinh \mu x$; $X(0) = a = 0$, $X(L) = b \sinh \mu L = 0 \Rightarrow b = 0$.

Step 3. Case lambda = 0

$X = a + bx$; $X(0) = a = 0$, $X(L) = bL = 0 \Rightarrow b = 0$.

Step 4. Case lambda > 0

$\lambda = \mu^2$: $X = a\cos \mu x + b \sin \mu x$; $X(0) = a = 0$; $b \sin \mu L = 0$ with $b \neq 0$ gives $\mu = n\pi/L$. Then $T'' + \omega_n^2 T = 0$ with $\omega_n = c\sqrt{\lambda_n} = \dfrac{n\pi c}{L}$, so $T = \alpha \cos \omega_n t + \beta \sin \omega_n t$. The condition $u(x,0) = X(x)T(0) = 0$ gives $\alpha = 0$. The normal modes are

$$u_n(x,t) = \sin \frac{n\pi x}{L} \, \sin \frac{n\pi c t}{L}, \qquad n = 1, 2, \ldots$$

KEY RESULTS

- $\dfrac{T''}{c^2 T} = \dfrac{X''}{X} = -\lambda$.
- $\lambda \le 0$: only $X \equiv 0$. $\lambda_n = \left(\dfrac{n\pi}{L}\right)^2$, $X_n = \sin\dfrac{n\pi x}{L}$.
- With $u(x,0) = 0$: $T_n = \sin \omega_n t$, $\omega_n = \dfrac{n\pi c}{L}$.

Part (b) · 1.5 points

Series solution with coefficients from the velocity

Step 1. Superposition and the velocity condition

$u = \sum_{n\ge 1} B_n \sin\dfrac{n\pi x}{L} \sin \omega_n t$ satisfies the equation, the fixed ends and $u(x,0) = 0$. Differentiating in t (chain rule: $\frac{d}{dt}\sin\omega_n t = \omega_n \cos\omega_n t$):

$$u_t(x,0) = \sum_{n\ge 1} \omega_n B_n \sin\frac{n\pi x}{L} = g(x) \;\Rightarrow\; \omega_n B_n = \frac{2}{L}\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx$$

Step 2. The integral

Let $k = n\pi/L$. First parts, $u = Lx - x^2$, $dv = \sin kx\,dx$, $du = (L-2x)dx$, $v = -\cos kx/k$:

$$\int_0^L (Lx-x^2)\sin kx\,dx = \Big[-\frac{(Lx-x^2)\cos kx}{k}\Big]_0^L + \frac{1}{k}\int_0^L (L-2x)\cos kx\,dx = \frac{1}{k}\int_0^L (L-2x)\cos kx\,dx$$

Second parts, $u = L-2x$, $dv = \cos kx\,dx$, $du = -2dx$, $v = \sin kx/k$:

$$\int_0^L (L-2x)\cos kx\,dx = \Big[\frac{(L-2x)\sin kx}{k}\Big]_0^L + \frac{2}{k}\int_0^L \sin kx\,dx = 0 + \frac{2}{k}\Big[-\frac{\cos kx}{k}\Big]_0^L = \frac{2(1-(-1)^n)}{k^2}$$

So $\displaystyle\int_0^L x(L-x)\sin kx\,dx = \frac{2(1-(-1)^n)}{k^3} = \frac{2L^3(1-(-1)^n)}{\pi^3 n^3}$.

Step 3. Coefficients

$$\omega_n B_n = \frac{2}{L}\cdot\frac{4V}{L^2}\cdot\frac{2L^3(1-(-1)^n)}{\pi^3 n^3} = \frac{16V(1-(-1)^n)}{\pi^3 n^3}$$

which is $\dfrac{32V}{\pi^3 n^3}$ for odd n and 0 for even n (the even modes have a node at the midpoint and are not excited by a velocity profile symmetric about it). Dividing by $\omega_n = n\pi c/L$:

$$B_n = \frac{32VL}{c\pi^4 n^4}\ (n \text{ odd}), \qquad u(x,t) = \frac{32VL}{c\pi^4}\sum_{n=1,3,5,\ldots}\frac{1}{n^4}\sin\frac{n\pi x}{L}\sin\frac{n\pi ct}{L}$$

KEY RESULTS

- $\dfrac{2}{L}\displaystyle\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx = \frac{32V}{\pi^3 n^3}$ for odd n, 0 for even n.
- $\omega_n B_n = \dfrac{32V}{\pi^3 n^3}$, so $B_n = \dfrac{32VL}{c\pi^4 n^4}$ for odd n.
- $u(x,t) = \dfrac{32VL}{c\pi^4}\displaystyle\sum_{n\ \text{odd}}\frac{1}{n^4}\sin\frac{n\pi x}{L}\sin\frac{n\pi ct}{L}$.

Part (c) · 1.25 points

d'Alembert check at the midpoint

Step 1. d'Alembert's formula

The general solution of $u_{tt} = c^2 u_{xx}$ is $F(x-ct) + H(x+ct)$; with $u(x,0)=0$ and $u_t(x,0) = G(x)$ it becomes $u(x,t) = \dfrac{1}{2c}\displaystyle\int_{x-ct}^{x+ct} G(s)\,ds$ (extension: the course solves the string by separation of variables; d'Alembert's form is used here only as an independent check).

Step 2. The integral at the midpoint

With $s = L/2 + \sigma$, $ds = d\sigma$, the limits $s = L/2 \mp ct$ become $\sigma = \mp ct$, and $s(L-s) = \left(\dfrac{L}{2}+\sigma\right)\left(\dfrac{L}{2}-\sigma\right) = \dfrac{L^2}{4} - \sigma^2$, so $g = V - \dfrac{4V\sigma^2}{L^2}$:

$$u\left(\frac{L}{2},t\right) = \frac{1}{2c}\int_{-ct}^{ct}\left(V - \frac{4V\sigma^2}{L^2}\right)d\sigma = \frac{1}{2c}\left[V\sigma - \frac{4V\sigma^3}{3L^2}\right]_{-ct}^{ct} = \frac{1}{2c}\left(2Vct - \frac{8Vc^3t^3}{3L^2}\right)$$

$$u\Big(\frac{L}{2},t\Big)=Vt-\frac{4Vc^2t^3}{3L^2},\qquad 0\le t\le\frac{L}{2c}$$

For small t, $u\approx Vt$: the midpoint starts with speed V, as imposed.

Step 3. Comparison with the series

At $t=L/(2c)$: d'Alembert gives $\frac{VL}{2c}-\frac{4Vc^2}{3L^2}\cdot\frac{L^3}{8c^3}=\frac{VL}{2c}-\frac{VL}{6c}=\frac{VL}{3c}$. In the series, $\sin\frac{n\pi}{2}\sin\frac{n\pi}{2}=\big((-1)^{(n-1)/2}\big)^2=1$ for odd n:

$$\frac{VL}{3c}=\frac{32VL}{c\pi^4}\sum_{n\text{ odd}}\frac{1}{n^4}\;\Rightarrow\;\sum_{n\text{ odd}}\frac{1}{n^4}=\frac{\pi^4}{96}$$

KEY RESULTS

- $u(L/2,t)=Vt-\dfrac{4Vc^2t^3}{3L^2}$ for $0\le t\le\dfrac{L}{2c}$.
- At $t=L/(2c)$: $u=\dfrac{VL}{3c}$, and the series gives $\dfrac{32VL}{c\pi^4}\sum_{n\text{ odd}}n^{-4}$.
- $\displaystyle\sum_{n\text{ odd}}\frac{1}{n^4}=\frac{\pi^4}{96}$.

Part (d) · 1 point

Energy of the modes and periodicity

Step 1. Energy of one mode

For $u_n=B_n\sin k_nx\,\sin\omega_nt$ ($k_n=n\pi/L$): $\partial_tu_n=B_n\omega_n\sin k_nx\cos\omega_nt$, $\partial_xu_n=B_nk_n\cos k_nx\sin\omega_nt$. With $\int_0^L\sin^2k_nx\,dx=\int_0^L\cos^2k_nx\,dx=L/2$ and $T_0=\rho c^2$:

$$E_n=\frac{\rho}{2}B_n^2\omega_n^2\frac{L}{2}\cos^2\omega_nt+\frac{\rho c^2}{2}B_n^2k_n^2\frac{L}{2}\sin^2\omega_nt=\frac{\rho L}{4}\omega_n^2B_n^2$$

since $c^2k_n^2=\omega_n^2$. With $\omega_nB_n=32V/(\pi^3n^3)$: $E_n=\dfrac{\rho L}{4}\cdot\dfrac{1024V^2}{\pi^6n^6}=\dfrac{256\rho LV^2}{\pi^6n^6}$.

Step 2. Total energy

E is conserved (the ends do no work), so $E = E(0) = \dfrac{\rho}{2}\displaystyle\int_0^L \frac{16V^2}{L^4}x^2(L-x)^2\,dx$. Expanding $x^2(L-x)^2 = L^2x^2 - 2Lx^3 + x^4$ and integrating: $\left[\dfrac{L^2x^3}{3} - \dfrac{Lx^4}{2} + \dfrac{x^5}{5}\right]_0^L = L^5\left(\dfrac{1}{3} - \dfrac{1}{2} + \dfrac{1}{5}\right) = \dfrac{L^5}{30}$. Hence $E = \dfrac{8\rho V^2}{L^4}\cdot\dfrac{L^5}{30} = \dfrac{4\rho L V^2}{15}$. The modes are orthogonal, so $E = \sum E_n$, and with $\sum_{n\ \text{odd}} n^{-6} = \pi^6/960$ the sum indeed gives $\dfrac{256}{960}\rho L V^2 = \dfrac{4}{15}\rho L V^2$.

Step 3. Fraction in the fundamental

$\dfrac{E_1}{E} = \dfrac{256/\pi^6}{4/15} = \dfrac{960}{\pi^6} \approx 0.99856$: a smooth, symmetric strike puts 99.86% of the energy in the fundamental; the string sounds an almost pure tone of frequency $c/(2L)$.

Step 4. Periodicity

$\omega_n = n\omega_1$, so every term has period $2\pi/\omega_1 = 2L/c$ and $u(x, t + 2L/c) = u(x,t)$. Only odd n appear, and $\sin(\omega_n t + n\pi) = -\sin\omega_n t$ for odd n, hence $u(x, t + L/c) = -u(x,t)$. At $t = L/c$: $u = 0$ and $u_t = \sum \omega_n B_n \sin k_n x \cos n\pi = -g(x)$: the string crosses its rest position with the velocity reversed. At $t = L/(2c)$: $\cos(n\pi/2) = 0$ for odd n, so $u_t = 0$: the string is momentarily at rest at its maximum displacement (midpoint $VL/(3c)$).

KEY RESULTS

- $E_n = \dfrac{\rho L}{4}\omega_n^2 B_n^2 = \dfrac{256\,\rho L V^2}{\pi^6 n^6}$ (n odd), constant in time.
- $E(0) = \dfrac{4\rho L V^2}{15}$, and $\sum_{n\ \text{odd}} E_n = \dfrac{256\rho L V^2}{\pi^6}\cdot\dfrac{\pi^6}{960} = \dfrac{4\rho L V^2}{15}$.
- Fraction in the fundamental: $\dfrac{960}{\pi^6} \approx 0.99856$.
- Period $\dfrac{2L}{c}$; at $t = L/(2c)$ the string is momentarily at rest with maximum displacement; at $t = L/c$ it crosses the rest position with velocity $-g(x)$, and $u(x, t + L/c) = -u(x,t)$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $c \to \infty$ (very stiff string): $u \to 0$, the displacement amplitude $VL/(3c)$ vanishes. Small t: $u \approx t\,g(x)$, consistent with the initial velocity. The coefficients decay as n^{-4}, faster than for a plucked string (n^{-2}), because the displacement is the time integral of a smooth velocity profile.
- **Checks:** dimensions: $B_n = \dfrac{32VL}{c\pi^4 n^4}$ is $\dfrac{(\mathrm{m/s})\,\mathrm{m}}{\mathrm{m/s}} = \mathrm{m}$; $E = \rho L V^2$ is (kg/m)(m)(m^2/s^2) = J. Two independent routes (series and d'Alembert) give the same midpoint displacement $VL/(3c)$.
- **Common mistakes in the exam:** (1) keeping $\cos\omega_n t$ terms although the initial displacement is zero; (2) setting B_n equal to the sine coefficients of g, forgetting the factor $1/\omega_n$ from differentiating in t; (3) using g itself instead of its odd periodic extension in d'Alembert's formula when $ct > L/2$; (4) giving the period as L/c: that is only the half-period after which u changes sign.